

Multistage Economic Dispatch Model with Piecewise Linear Decision Rules

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Goal:

- Implement a piecewise linear decision rule multistage economic dispatch problem with stochastic demand.

Benchmark

- Perfect information model;
- Linear decision rule model from [1]

Project goal

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Multistage day-ahead scheduling of energy and reserves[☆]

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ABSTRACT

This paper presents a new economic dispatch model for the day-ahead scheduling of energy and reserves under uncertainty. The proposed model considers a multistage stochastic framework whereby scheduling decisions are dynamically updated according to observed new information about uncertain parameters such as nodal net injections. As a result, the proposed model provides a one-hour-ahead implementable energy and reserve schedule. Moreover, a particularization of this multistage model is presented, namely a two-stage model, in which a single generation and reserve schedule is provided for each hour of the day. A recently developed regularized linear decision rules framework is used to reduce the computational complexity of the multistage stochastic linear problem at hand and to prevent the in-sample overfitting issue and the threat of poor out-of-sample performance. Numerical simulations based on the IEEE 300-bus system demonstrate the effectiveness of the proposed approach, as well as its economic and operational advantages over the widely used two-stage model.

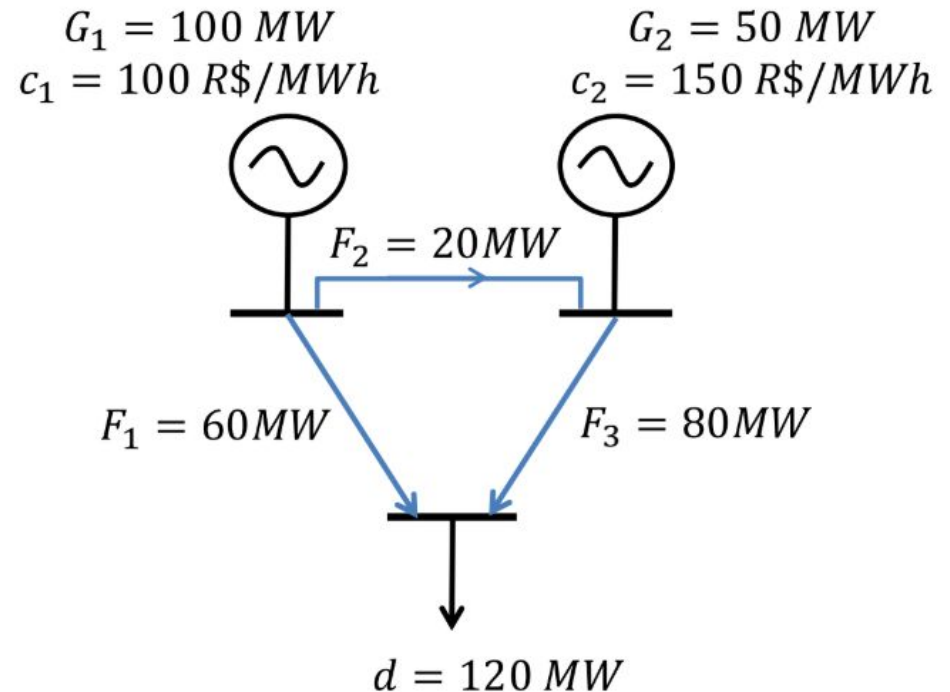
Economic Dispatch Problem

Objective:

- Attend demand with minimal cost while respecting electrical, flow and capacity constraints

Decision variables:

- Generations g ;
- Flows f ;
- Phase θ
- Load shedding δ ;
- Generation curtailment γ



$$f_l = \frac{1}{x_l} (\theta_{ori(l)} - \theta_{des(l)})$$

Economic Dispatch Problem

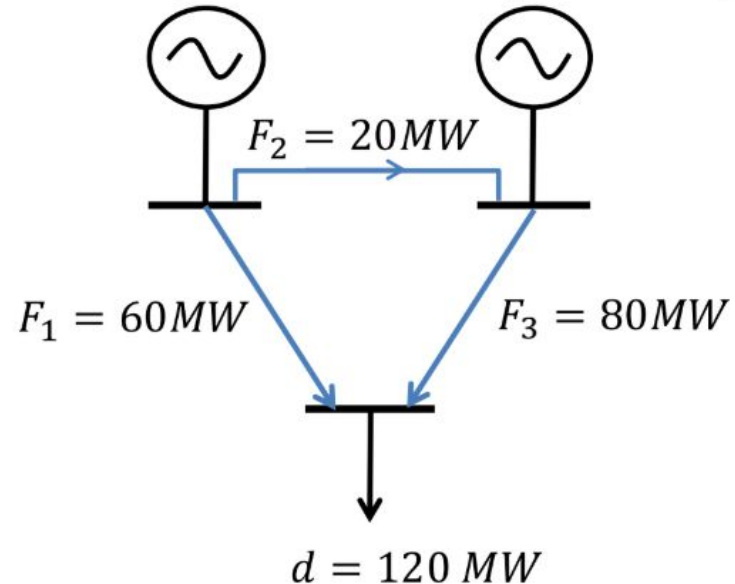
$$\min_{g_i \geq 0, f_l} 100g_1 + 150g_2$$

s.a:

$$\begin{array}{rclcl} g_1 & -f_1 - f_2 & = & 0 \\ g_2 & +f_2 - f_3 & = & 0 \\ & f_1 + f_3 & = & d \\ -60 \leq & f_1 & \leq & 60 \\ -20 \leq & f_2 & \leq & 20 \\ -80 \leq & f_3 & \leq & 80 \\ g_1 & & \leq & 100 \\ g_2 & & \leq & 50 \\ & f_1 = \theta_1 - \theta_3 \\ & f_2 = \theta_1 - \theta_2 \\ & f_3 = \theta_2 - \theta_3 \end{array}$$

$$G_1 = 100 \text{ MW} \\ c_1 = 100 \text{ R\$/MWh}$$

$$G_2 = 50 \text{ MW} \\ c_2 = 150 \text{ R\$/MWh}$$



$$f_l = \frac{1}{x_l} (\theta_{ori(l)} - \theta_{des(l)})$$

IEEE 300-bus System

The system is composed of

- 300 Buses
- 441 Lines
- 69 Generators
- 191 Loads

System parameters

- Curtailment cost:
- Load shedding cost
- Decision rules slacks cost
- All other parameters were used as given.

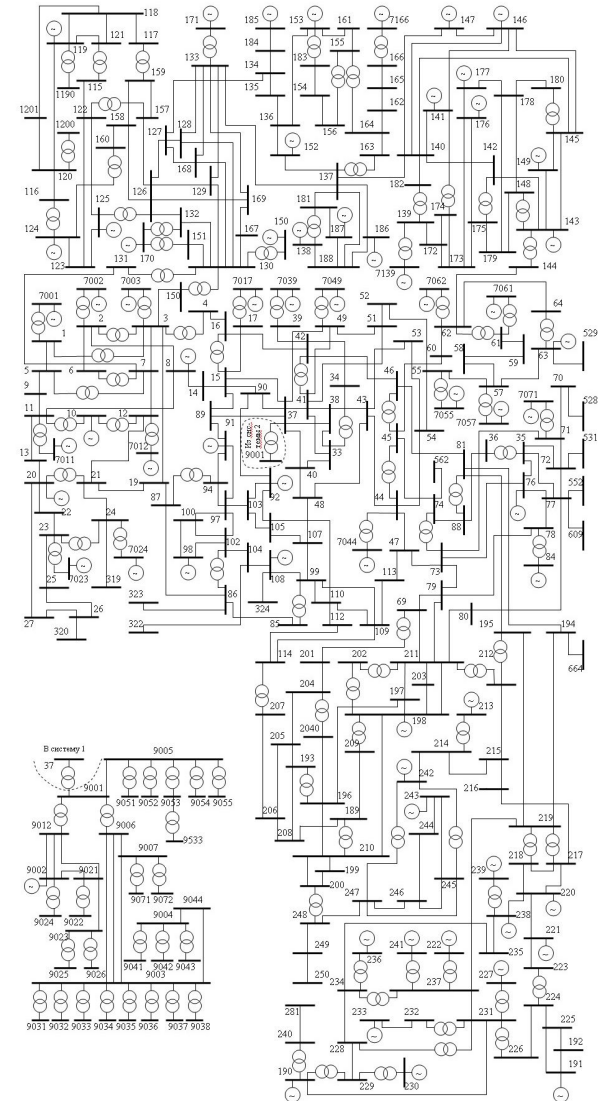


Рис.1. IEEE тестовая схема, состоящая из 300 узлов

* Taken from <https://al-roomi.org/power-flow/300-bus-system>

Objective Function:

$$\begin{aligned}
 \text{Min}_{\substack{(g, r^{up}, r^{dn}, \Delta)_{t, \omega, i}, \\ (f, f^{RT})_{t, \omega, j}, \\ (\beta^{(g)}, \beta^{(up)}, \beta^{(dn)})_{t, i, 0}, \\ (\beta^{(g)}, \beta^{(up)}, \beta^{(dn)})_{t, i, b, l}, \\ (\delta, \delta^{RT}, \gamma, \gamma^{RT}, \theta, \theta^{RT})_{t, \omega, b}, \\ (\Phi^{(g)}, \Phi^{(up)}, \Phi^{(dn)})_{t, i, b, l}}} & \sum_{\omega \in \Omega} P_{\omega} \sum_{i \in \mathcal{G}} \sum_{t \in T} \left[c_i^g g_{i, t, \omega} + c_i^{up} r_{i, t, \omega}^{up} + c_i^{dn} r_{i, t, \omega}^{dn} \right] \\
 & + \sum_{b \in \mathcal{B}} \left[c_b^{\delta} (\delta_{b, t, \omega} + \delta_{b, t, \omega}^{RT}) + c_b^{\gamma} (\gamma_{b, t, \omega} + \gamma_{b, t, \omega}^{RT}) \right] \\
 & + \lambda \sum_{t \in T} \sum_{i \in \mathcal{G}} \sum_{b \in \mathcal{B}} \sum_{\ell \in \mathcal{L}} \left(\Phi_{i, t, b, \ell}^g + \Phi_{i, t, b, \ell}^{up} + \Phi_{i, t, b, \ell}^{dn} \right)
 \end{aligned}$$

Power balance and demand constraint :

$$\sum_{i \in U_b} g_{t,\omega,i} + \sum_{j \in L_b^+} f_{t,\omega,j} - \sum_{j \in L_b^-} f_{t,\omega,j} + \delta_{t,\omega,b} - \gamma_{t,\omega,b} = \hat{d}_{t,\omega,b}, \quad \forall t, \omega, b \quad (2)$$

$$\sum_{i \in U_b} (g_{t,\omega,i} + \Delta_{t,\omega,i}) + \sum_{j \in L_b^+} f_{t,\omega,j}^{RT} - \sum_{j \in L_b^-} f_{t,\omega,j}^{RT} + \delta_{t,\omega,b}^{RT} - \gamma_{t,\omega,b}^{RT} = d_{t,\omega,b}, \quad \forall t, \omega, b \quad (3)$$

Second Kirchhoff's law and flows' bounds:

$$f_{t,\omega,j} = \frac{\theta_{t,\omega,b_j^+} - \theta_{t,\omega,b_j^-}}{x_j}, \quad \forall t, \omega, j \quad (4)$$

$$f_{t,\omega,j}^{RT} = \frac{\theta_{t,\omega,b_j^+}^{RT} - \theta_{t,\omega,b_j^-}^{RT}}{x_j}, \quad \forall t, \omega, j \quad (5)$$

$$-F_j \leq f_{t,\omega,j} \leq F_j, \quad \forall t, \omega, j \quad (6)$$

$$-F_j \leq f_{t,\omega,j}^{RT} \leq F_j, \quad \forall t, \omega, j \quad (7)$$

Generation Limits:

$$\underline{G}_i \leq g_{t,\omega,i} \leq \bar{G}_i \quad \forall t, \omega, i \quad (8)$$

$$g_{t,\omega,i} + r_{t,\omega,i}^{up} \leq \bar{G}_i \quad \forall t, \omega, i \quad (9)$$

$$g_{t,\omega,i} - r_{t,\omega,i}^{dn} \geq \underline{G}_i \quad \forall t, \omega, i \quad (10)$$

Ramp, redispatch and reserve bounds :

$$-RD_i \leq g_{t,\omega,i} + \Delta_{t,\omega,i} - g_{t-1,\omega,i} - \Delta_{t-1,\omega,i} \leq RU_i, \quad \forall t, \omega, i \quad (11)$$

$$-r_{t,\omega,i}^{dn} \leq \Delta_{t,\omega,i} \leq r_{t,\omega,i}^{up}, \quad \forall t, \omega, i \quad (12)$$

$$0 \leq r_{t,\omega,i}^{up} \leq R_i^{up}, \quad \forall t, \omega, i \quad (13)$$

$$0 \leq r_{t,\omega,i}^{dn} \leq R_i^{dn}, \quad \forall t, \omega, i \quad (14)$$

Load-shedding and Curtailment :

$$0 \leq \delta_{t,\omega,b} \leq \hat{d}_{t,\omega,b}, \quad \forall t, \omega, b \quad (15)$$

$$0 \leq \delta_{t,\omega,b}^{RT} \leq d_{t,\omega,b}, \quad \forall t, \omega, b \quad (16)$$

$$0 \leq \gamma_{t,\omega,b} \leq \bar{G}_b, \quad \forall t, \omega, b \quad (17)$$

$$0 \leq \gamma_{t,\omega,b}^{RT} \leq \bar{G}_b, \quad \forall t, \omega, b \quad (18)$$

Load-shedding and Curtailment :

$$0 \leq \delta_{t,\omega,b} \leq \hat{d}_{t,\omega,b}, \quad \forall t, \omega, b \quad (15)$$

$$0 \leq \delta_{t,\omega,b}^{RT} \leq d_{t,\omega,b}, \quad \forall t, \omega, b \quad (16)$$

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$$0 \leq \gamma_{t,\omega,b}^{RT} \leq \bar{G}_b, \quad \forall t, \omega, b \quad (18)$$

LDR Expressions:

$$g_{t,\omega,i} = \beta_{t,i,0}^{(g)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \beta_{t,i,b,l}^{(g)} d_{t-l,\omega,b} \quad \forall t, \omega, i \quad (19)$$

$$r_{t,\omega,i}^{up} = \beta_{t,i,0}^{(up)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \beta_{t,i,b,l}^{(up)} d_{t-l,\omega,b} \quad \forall t, \omega, i \quad (20)$$

$$r_{t,\omega,i}^{dn} = \beta_{t,i,0}^{(dn)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \beta_{t,i,b,l}^{(dn)} d_{t-l,\omega,b} \quad \forall t, \omega, i \quad (20)$$

$$\Phi_{t,i,b,l}^{(g)} - \beta_{t,i,b,l}^{(g)} \geq 0 \quad \forall t, i, b, l \quad (21)$$

$$\Phi_{t,i,b,l}^{(g)} + \beta_{t,i,b,l}^{(g)} \geq 0 \quad \forall t, i, b, l \quad (22)$$

$$\Phi_{t,i,b,l}^{(up)} - \beta_{t,i,b,l}^{(up)} \geq 0 \quad \forall t, i, b, l \quad (23)$$

$$\Phi_{t,i,b,l}^{(up)} + \beta_{t,i,b,l}^{(up)} \geq 0 \quad \forall t, i, b, l \quad (24)$$

$$\Phi_{t,i,b,l}^{(dn)} - \beta_{t,i,b,l}^{(dn)} \geq 0 \quad \forall t, i, b, l \quad (25)$$

$$\Phi_{t,i,b,l}^{(dn)} + \beta_{t,i,b,l}^{(dn)} \geq 0 \quad \forall t, i, b, l \quad (26)$$

LDR Objective Function:

$$\begin{aligned}
 \text{Min}_{\substack{(g, r^{up}, r^{dn}, \Delta)_{t, \omega, i}, \\ (f, f^{RT})_{t, \omega, j}, \\ (\beta^{(g)}, \beta^{(up)}, \beta^{(dn)})_{t, i, 0}, \\ (\beta^{(g)}, \beta^{(up)}, \beta^{(dn)})_{t, i, b, l}, \\ (\delta, \delta^{RT}, \gamma, \gamma^{RT}, \theta, \theta^{RT})_{t, \omega, b}, \\ (\Phi^{(g)}, \Phi^{(up)}, \Phi^{(dn)})_{t, i, b, l}}} \quad & \sum_{\omega \in \Omega} P_{\omega} \sum_{i \in \mathcal{G}} \sum_{t \in T} \left[c_i^g g_{i, t, \omega} + c_i^{up} r_{i, t, \omega}^{up} + c_i^{dn} r_{i, t, \omega}^{dn} \right] \\
 & + \sum_{b \in \mathcal{B}} \left[c_b^{\delta} (\delta_{b, t, \omega} + \delta_{b, t, \omega}^{RT}) + c_b^{\gamma} (\gamma_{b, t, \omega} + \gamma_{b, t, \omega}^{RT}) \right] \\
 & + \lambda \sum_{t \in T} \sum_{i \in \mathcal{G}} \sum_{b \in \mathcal{B}} \sum_{\ell \in \mathcal{L}} \left(\Phi_{i, t, b, \ell}^g + \Phi_{i, t, b, \ell}^{up} + \Phi_{i, t, b, \ell}^{dn} \right)
 \end{aligned}$$

PWL Objective Function:

$$\begin{aligned}
 \text{Min}_{\substack{(g, r^{up}, r^{dn}, \Delta)_{t, \omega, i}, \\ (f, f^{RT})_{t, \omega, j}, \\ (\beta^{(g)}, \beta^{(up)}, \beta^{(dn)})_{t, i, 0}, \\ (\beta^{(g)}, \beta^{(up)}, \beta^{(dn)})_{t, i, b, l, r}, \\ (\delta, \delta^{RT}, \gamma, \gamma^{RT}, \theta, \theta^{RT})_{t, \omega, b}, \\ (\Phi^{(g)}, \Phi^{(up)}, \Phi^{(dn)})_{t, i, b, l, r}}} \quad & \sum_{\omega \in \Omega} P_{\omega} \sum_{i \in \mathcal{G}} \sum_{t \in T} \left[c_i^g g_{i, t, \omega} + c_i^{up} r_{i, t, \omega}^{up} + c_i^{dn} r_{i, t, \omega}^{dn} \right] \\
 & + \sum_{b \in \mathcal{B}} \left[c_b^{\delta} (\delta_{b, t, \omega} + \delta_{b, t, \omega}^{RT}) + c_b^{\gamma} (\gamma_{b, t, \omega} + \gamma_{b, t, \omega}^{RT}) \right] \\
 & + \lambda \sum_{t \in T} \sum_{i \in \mathcal{G}} \sum_{b \in \mathcal{B}} \sum_{\ell \in \mathcal{L}} \sum_{r \in \mathcal{R}} \left(\Phi_{i, t, b, \ell, r}^g + \Phi_{i, t, b, \ell, r}^{up} + \Phi_{i, t, b, \ell, r}^{dn} \right)
 \end{aligned}$$

LDR Expressions:

$$g_{t,\omega,i} = \beta_{t,i,0}^{(g)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \beta_{t,i,b,l}^{(g)} d_{t-l,\omega,b} \quad \forall t, \omega, i \quad (19)$$

$$r_{t,\omega,i}^{up} = \beta_{t,i,0}^{(up)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \beta_{t,i,b,l}^{(up)} d_{t-l,\omega,b} \quad \forall t, \omega, i \quad (20)$$

$$r_{t,\omega,i}^{dn} = \beta_{t,i,0}^{(dn)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \beta_{t,i,b,l}^{(dn)} d_{t-l,\omega,b} \quad \forall t, \omega, i \quad (20)$$

$$\Phi_{t,i,b,l}^{(g)} - \beta_{t,i,b,l}^{(g)} \geq 0 \quad \forall t, i, b, l \quad (21)$$

$$\Phi_{t,i,b,l}^{(g)} + \beta_{t,i,b,l}^{(g)} \geq 0 \quad \forall t, i, b, l \quad (22)$$

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$$\Phi_{t,i,b,l}^{(dn)} - \beta_{t,i,b,l}^{(dn)} \geq 0 \quad \forall t, i, b, l \quad (25)$$

$$\Phi_{t,i,b,l}^{(dn)} + \beta_{t,i,b,l}^{(dn)} \geq 0 \quad \forall t, i, b, l \quad (26)$$

PWL Expressions:

$$g_{t,\omega,i} = \beta_{t,i,0}^{(g)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \sum_{r \in \mathcal{R}} \beta_{t,i,b,l,r}^{(g)} d_{t-l,\omega,b,r}^{lift} \quad \forall t, \omega, i \quad (19)$$

$$r_{t,\omega,i}^{up} = \beta_{t,i,0}^{(up)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \sum_{r \in \mathcal{R}} \beta_{t,i,b,l,r}^{(up)} d_{t-l,\omega,b,r}^{lift} \quad \forall t, \omega, i \quad (20)$$

$$r_{t,\omega,i}^{dn} = \beta_{t,i,0}^{(dn)} + \sum_{b \in \mathcal{B}} \sum_{l \in \mathcal{L}} \sum_{r \in \mathcal{R}} \beta_{t,i,b,l,r}^{(dn)} d_{t-l,\omega,b,r}^{lift} \quad \forall t, \omega, i \quad (20)$$

$$\Phi_{t,i,b,l,r}^{(g)} - \beta_{t,i,b,l,r}^{(g)} \geq 0 \quad \forall t, i, b, l, r \quad (21)$$

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$$\Phi_{t,i,b,l,r}^{(up)} - \beta_{t,i,b,l,r}^{(up)} \geq 0 \quad \forall t, i, b, l, r \quad (23)$$

$$\Phi_{t,i,b,l,r}^{(up)} + \beta_{t,i,b,l,r}^{(up)} \geq 0 \quad \forall t, i, b, l, r \quad (24)$$

$$\Phi_{t,i,b,l,r}^{(dn)} - \beta_{t,i,b,l,r}^{(dn)} \geq 0 \quad \forall t, i, b, l, r \quad (25)$$

$$\Phi_{t,i,b,l,r}^{(dn)} + \beta_{t,i,b,l,r}^{(dn)} \geq 0 \quad \forall t, i, b, l, r \quad (26)$$

The demand was divided into $R = 5$ uniformly distributed partitions, where Z_r represents the upper bound of each partition. Then, the lifted demand was created using the following function:

$$d_r^{\text{lift}}(d) = \begin{cases} \min(d, Z_r), & \text{if } r = 1, \\ \max(\min(d, Z_r) - Z_{r-1}, 0), & \text{if } 1 < r < 5, \\ \max(d - Z_{r-1}, 0), & \text{if } r = 5. \end{cases}$$

Original methodology from [1]:

- Demand simulations from autoregressive process

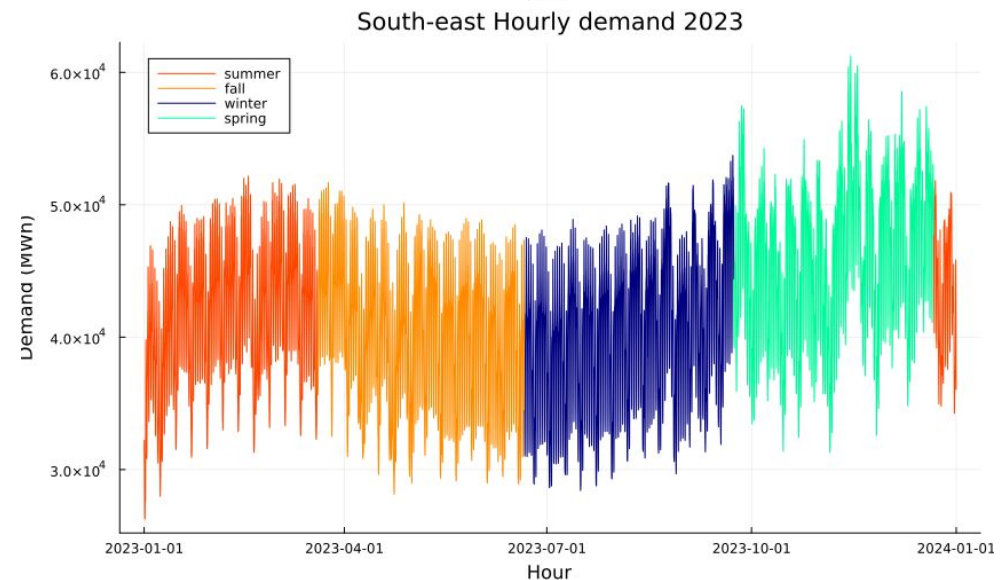
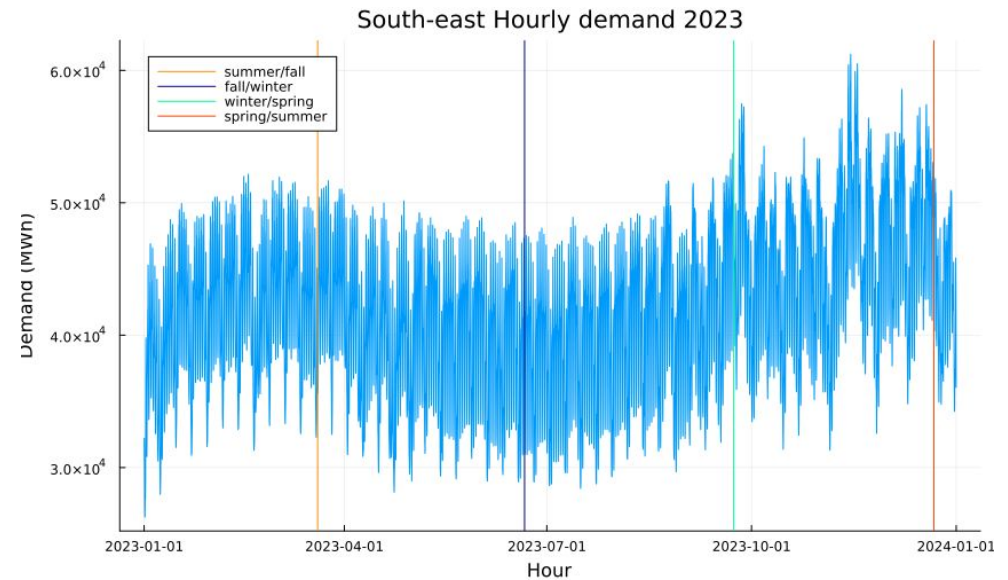
Proposed methodology:

- Demand simulations from a typical day, computed from real demand data and divided among the system buses.

Demand Scenarios

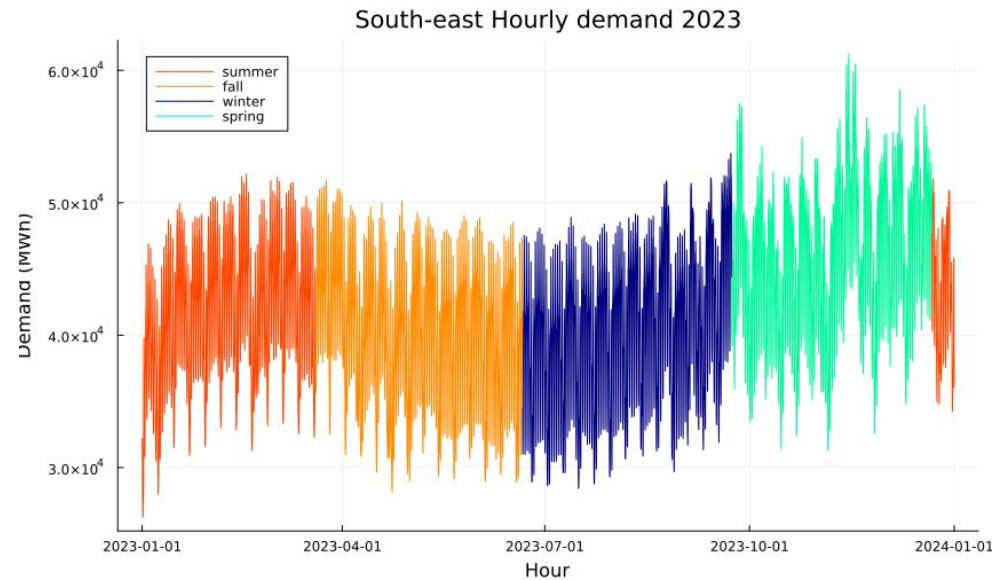
2023 South-east hourly demand time series

Divided into 4 seasons in order to consider similar days

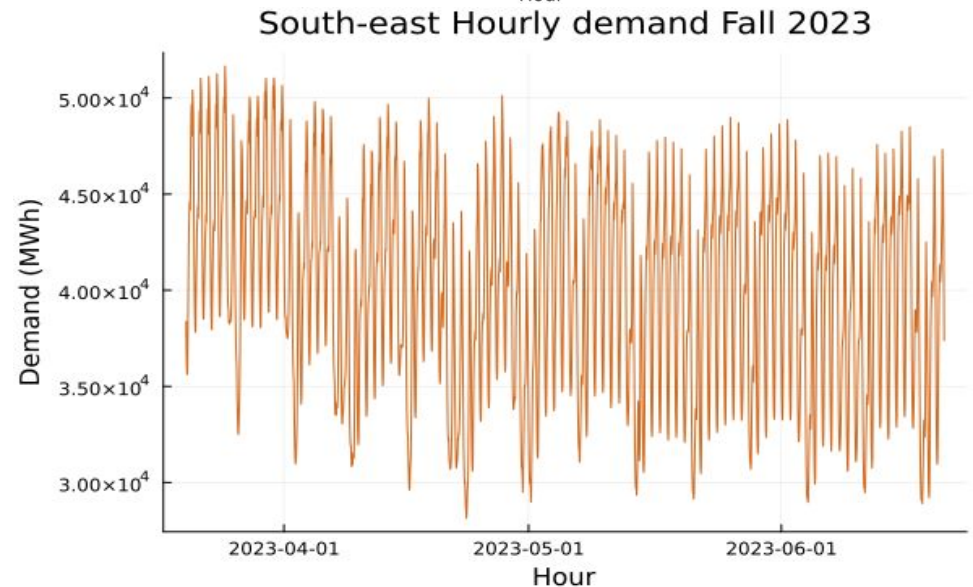


Demand Scenarios

Divided into 4 seasons in order
to consider similar days

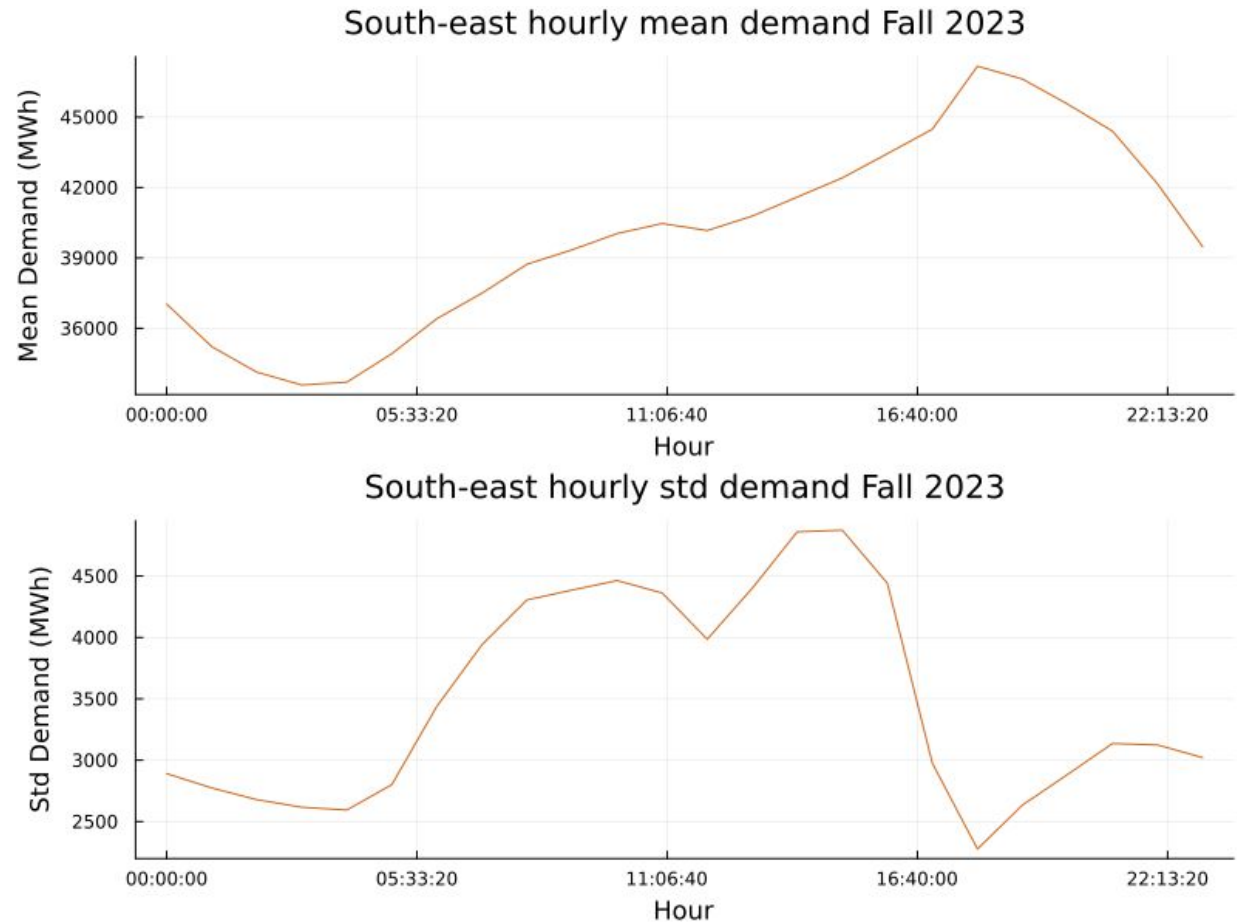


The fall season was chosen



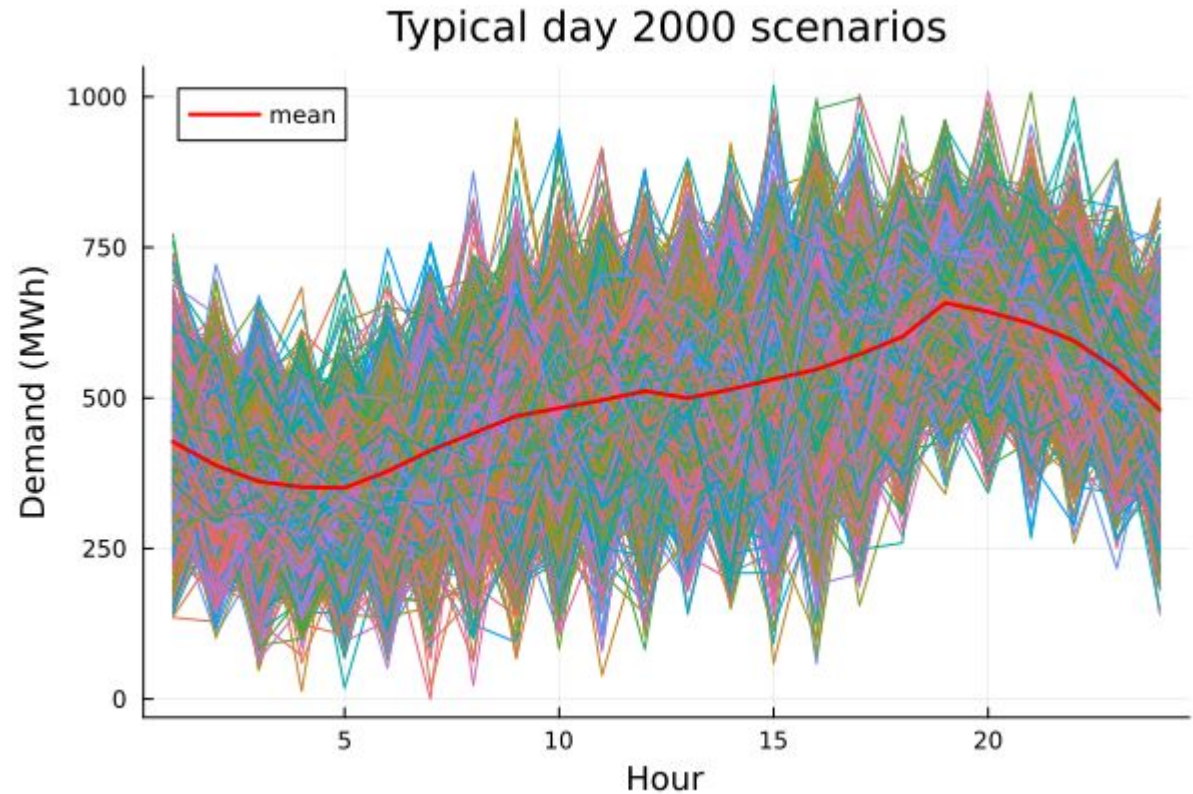
Demand Scenarios

Computed hourly mean
and standard deviation



Demand Scenarios

Simulated scenarios
from the fitted mean
and standard
deviations



Demand scenarios adjustments:

- Scaled to fit into the system parameters

$$\hat{d}_{\text{scaled}} = \left(\frac{\hat{d} - d_{\min}}{d_{\max} - d_{\min}} \right) \cdot \left(d_{\max}^{(0)} - d_{\min}^{(0)} \right) + d_{\min}^{(0)}$$

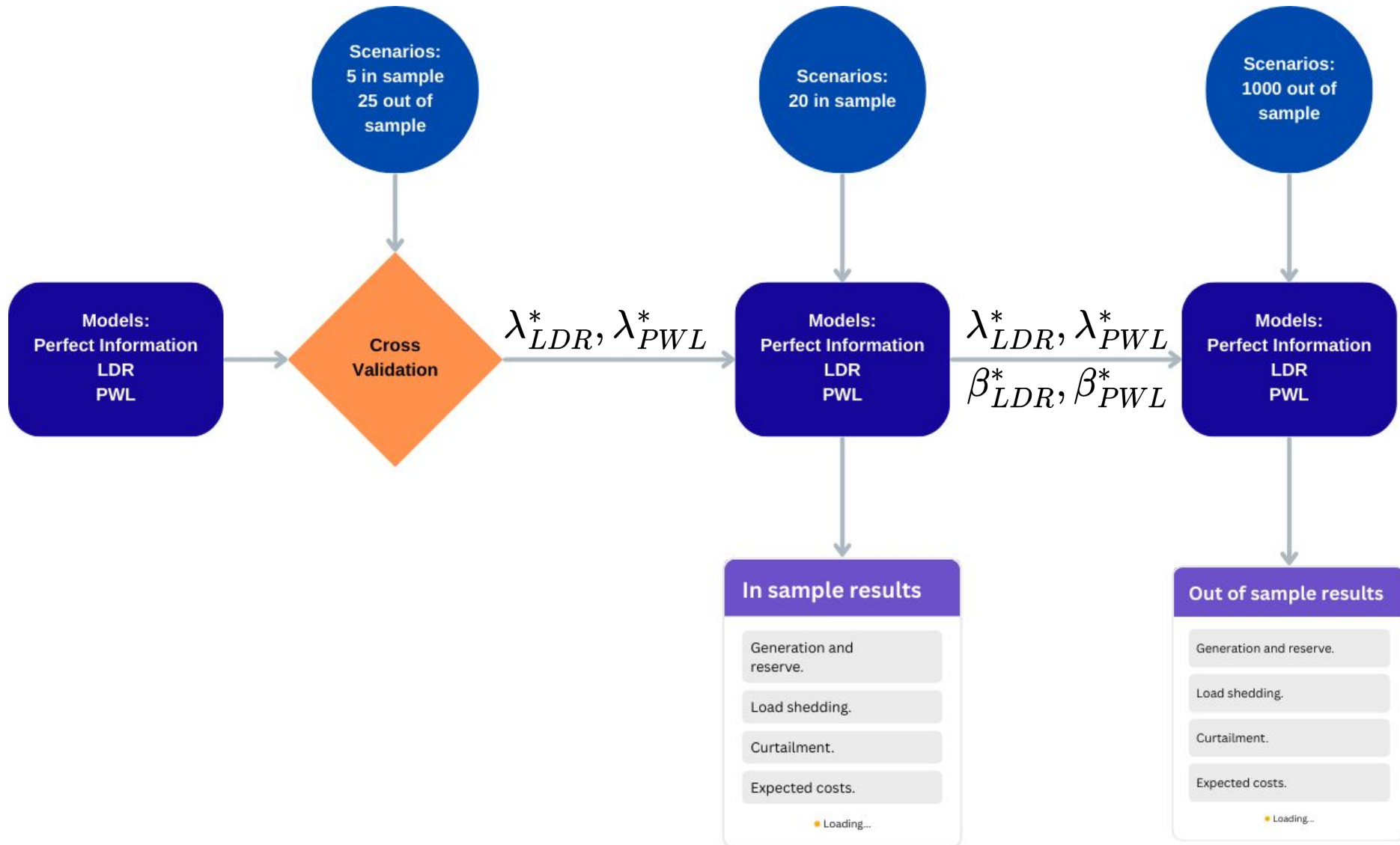
- Divided into the buses proportionally to the bus initial demand from data.

$$\hat{d}_b = \frac{d_b^{(0)}}{\sum_b d_b^{(0)}} \cdot \hat{d}_{\text{scaled}}$$

We have performed the following experiments:

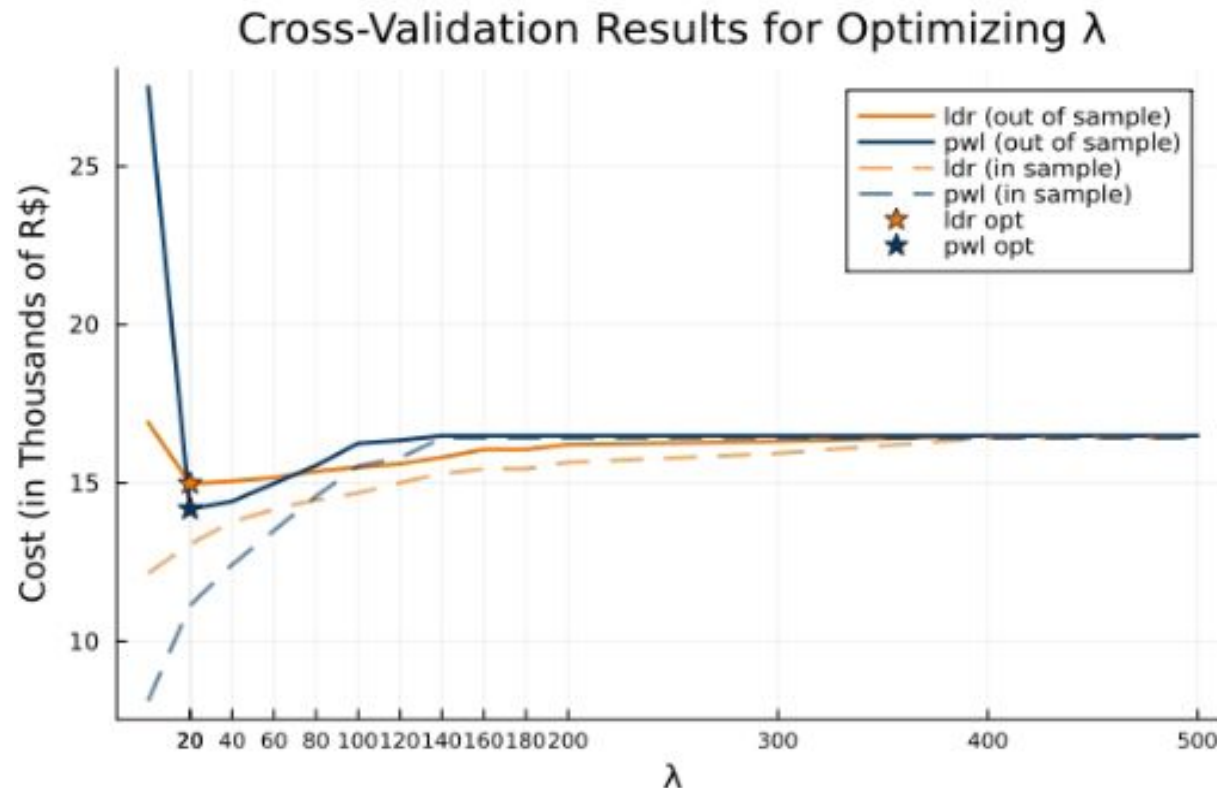
1. **Cross validation** to determine the best value for the decision rule regularization parameter based on out of sample performance
 - a. 5 demand scenarios for in sample models
 - b. 25 demand scenarios for out of sample models
2. **In sample models**
 - a. 20 new demand scenarios
 - b. Optimal regularization parameter
3. **Out of sample models**
 - a. 1000 new demand scenarios
 - b. Optimal regularization parameter
 - c. Decision rules coefficients from in sample models

Experiments



Cross Validation Decision Rule Regularization

- λ in $\{0, 20, 40, \dots, 180, 200, 300, 400, \dots, 1000\}$
- Out of sample optimal value: $\lambda = 20$



	In Sample			Out of Sample		
	PI	LDR	PWL	PI	LDR	PWL
Total	72137.186	172268.291	158945.851	72618.031	94311.825	90735.68
Generation	67037.224	93825.746	91236.824	68380.677	71074.327	70127.62
Reserve Up	4460.609	7183.164	7465.502	3805.654	7115.974	7399.561
Reserve Down	639.353	11101.952	9959.629	431.7	11095.113	9962.139
Curtailement	0.0	50113.779	45121.415	0.0	5026.411	3246.36
RT Curtailement	0.0	3430.734	4201.461	0.0	0.0	0.0
Load Shedding	0.0	6612.916	961.02	0.0	0.0	0.0
RT Load Shedding	0.0	0.0	0.0	0.0	0.0	0.0

Table 1: Costs results.

	In Sample			Out of Sample		
	PI	LDR	PWL	PI	LDR	PWL
Generation	10431.06	14442.373	14061.916	10678.686	11083.565	10940.181
Redispatch	1928.669	-1806.298	-1363.758	1688.846	1283.967	1427.351
Reserve Up	2264.45	3552.967	3693.886	1915.57	3514.159	3655.374
Reserve Down	335.781	5829.374	5230.688	226.724	5826.272	5232.006
Curtailement	0.0	4036.678	3634.542	0.0	404.879	261.495
RT Curtailement	0.0	276.347	338.429	0.0	0.0	0.0
Load Shedding	0.0	25.365	3.686	0.0	0.0	0.0
RT Load Shedding	0.0	0.0	0.0	0.0	0.0	0.0
Model Creation Time	7.094	69.189	386.173	4.254	33.946	210.275
Optimization Time	22.943	1891.767	21427.211	23.022	37.226	91.31

Table 2: Decision variables results.

	In Sample			Out of Sample		
	PI	LDR	PWL	PI	LDR	PWL
Total	72137.186	172268.291	158945.851	72618.031	94311.825	90735.68
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Table 1: Costs results.

Results

	In Sample			Out of Sample		
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Load Shedding	0.0	6612.916	961.02	0.0	0.0	0.0
RT Load Shedding	0.0	0.0	0.0	0.0	0.0	0.0

Table 1: Costs results.

Results

	In Sample			Out of Sample		
	PI	LDR	PWL	PI	LDR	PWL
Total	72137.186	172268.291	158945.851	72618.031	94311.825	90735.68
Generation	67037.224	93825.746	91236.824	68380.677	71074.327	70127.62
Reserve Up	4460.609	7183.164	7465.502	3805.654	7115.974	7399.561
Reserve Down	639.353	11101.952	9959.629	431.7	11095.113	9962.139
Curtailement	0.0	50113.779	45121.415	0.0	5026.411	3246.36
RT Curtailement	0.0	3430.734	4201.461	0.0	0.0	0.0
Load Shedding	0.0	6612.916	961.02	0.0	0.0	0.0
RT Load Shedding	0.0	0.0	0.0	0.0	0.0	0.0

Table 1: Costs results.

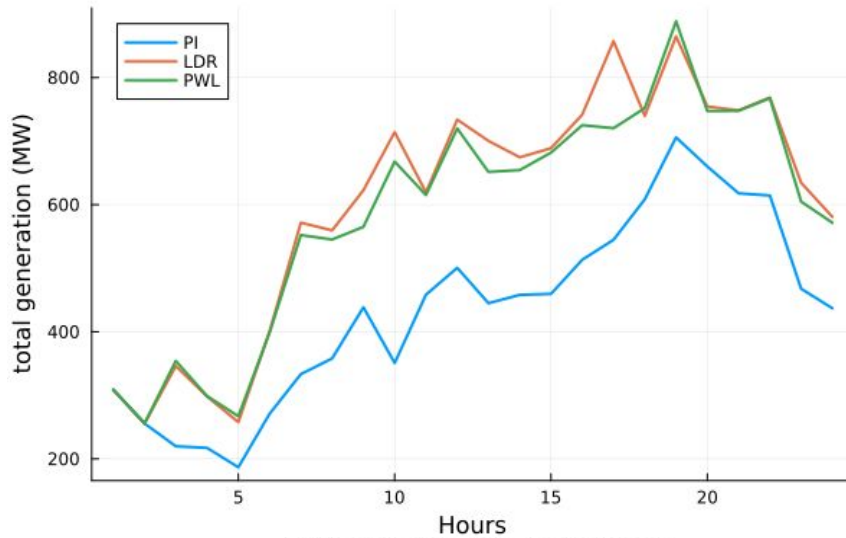
Results

	In Sample			Out of Sample		
	PI	LDR	PWL	PI	LDR	PWL
Generation	10431.06	14442.373	14061.916	10678.686	11083.565	10940.181
Redispatch	1928.669	-1806.298	-1363.758	1688.846	1283.967	1427.351
Reserve Up	2264.45	3552.967	3693.886	1915.57	3514.159	3655.374
Reserve Down	335.781	5829.374	5230.688	226.724	5826.272	5232.006
Curtailement	0.0	4036.678	3634.542	0.0	404.879	261.495
RT Curtailement	0.0	276.347	338.429	0.0	0.0	0.0
Load Shedding	0.0	25.365	3.686	0.0	0.0	0.0
RT Load Shedding	0.0	0.0	0.0	0.0	0.0	0.0
Model Creation Time	7.094	69.189	386.173	4.254	33.946	210.275
Optimization Time	22.943	1891.767	21427.211	23.022	37.226	91.31

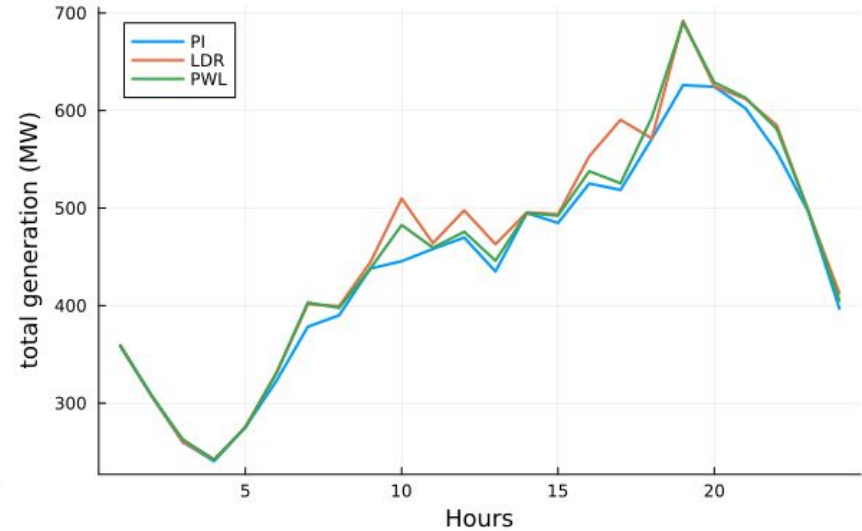
Table 2: Decision variables results.

Results

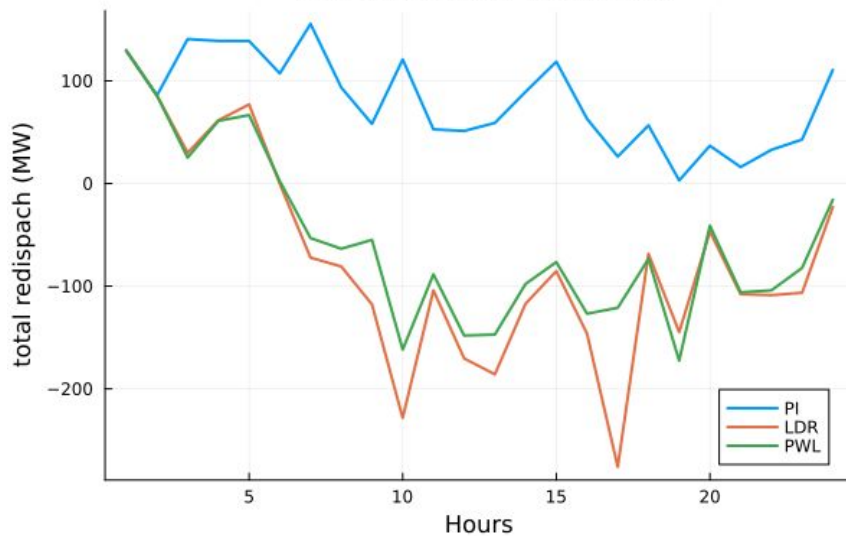
Total in sample generation



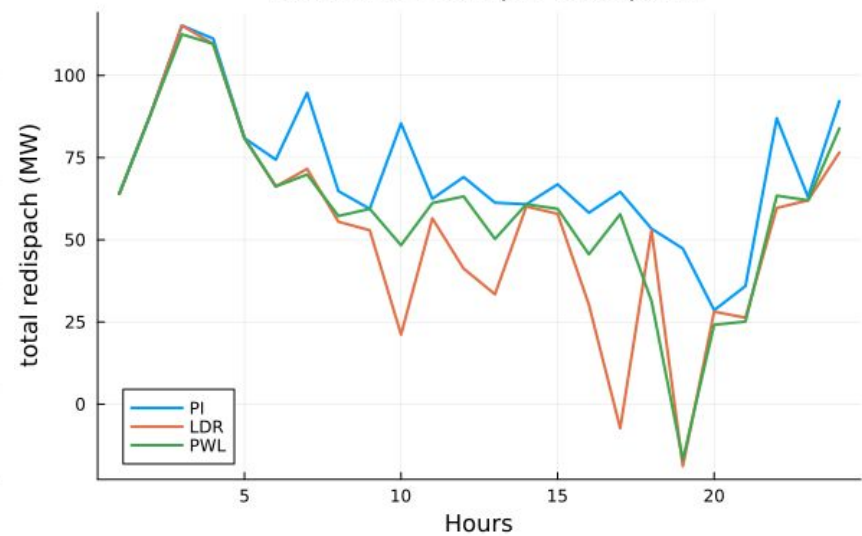
Total out of sample generation



Total in sample redispatch

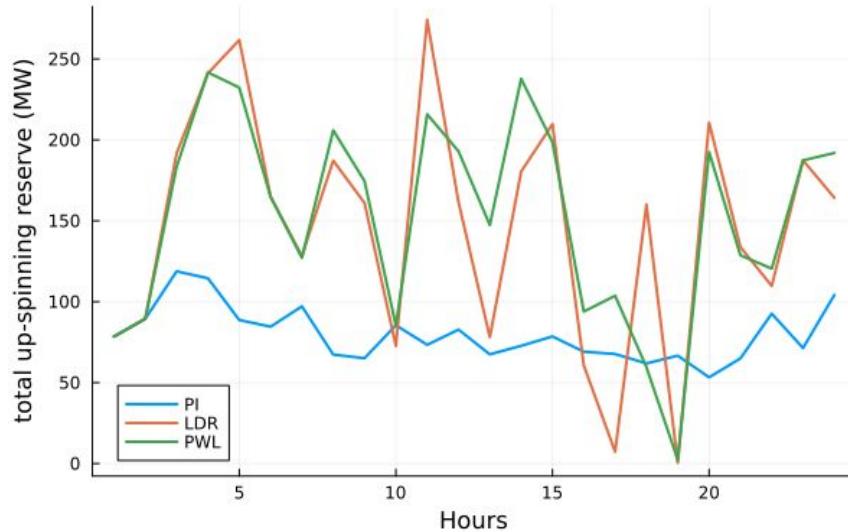


Total out of sample redispatch

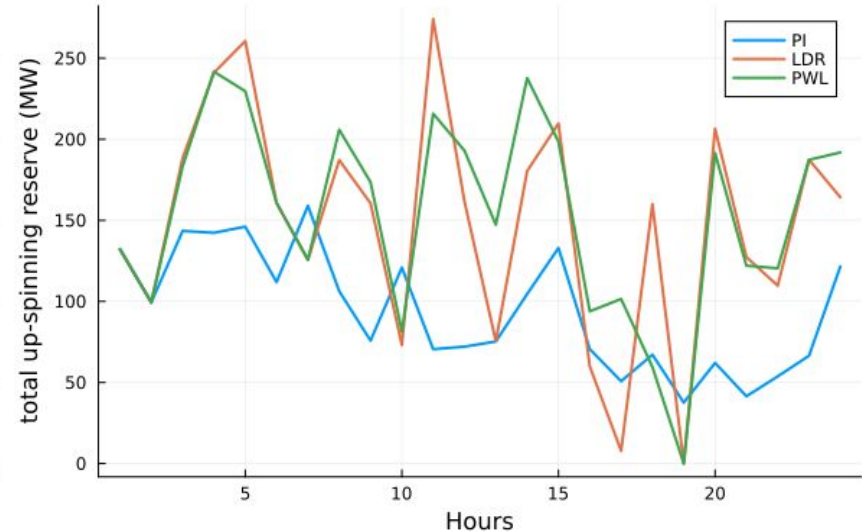


Results

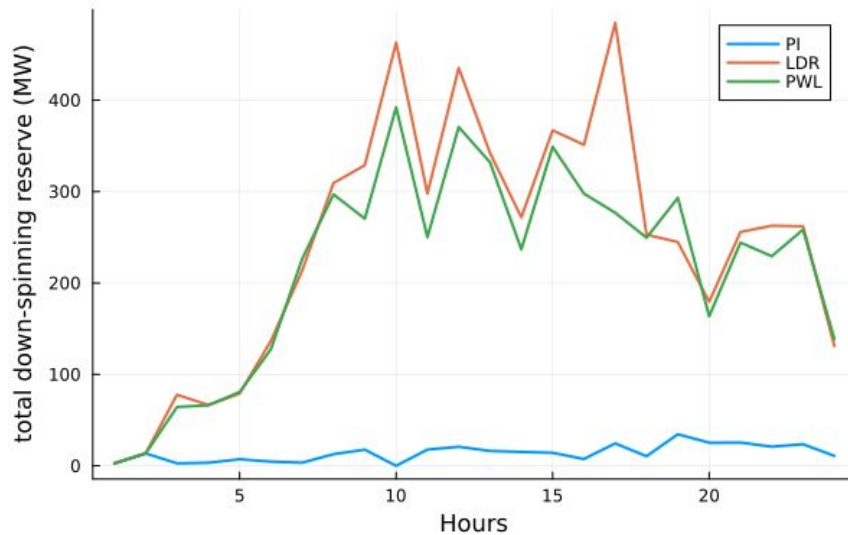
Total out of sample up-spinning reserve



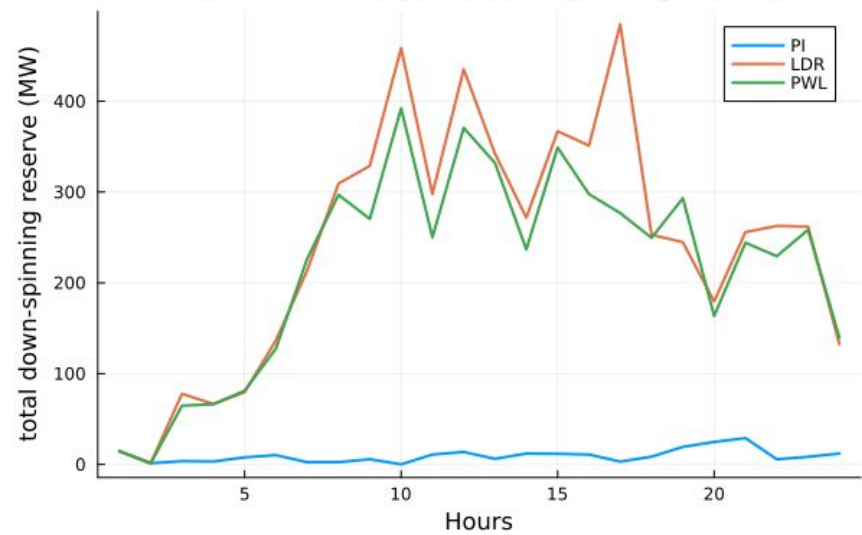
Total in sample up-spinning reserve



Total in sample down-spinning reserve

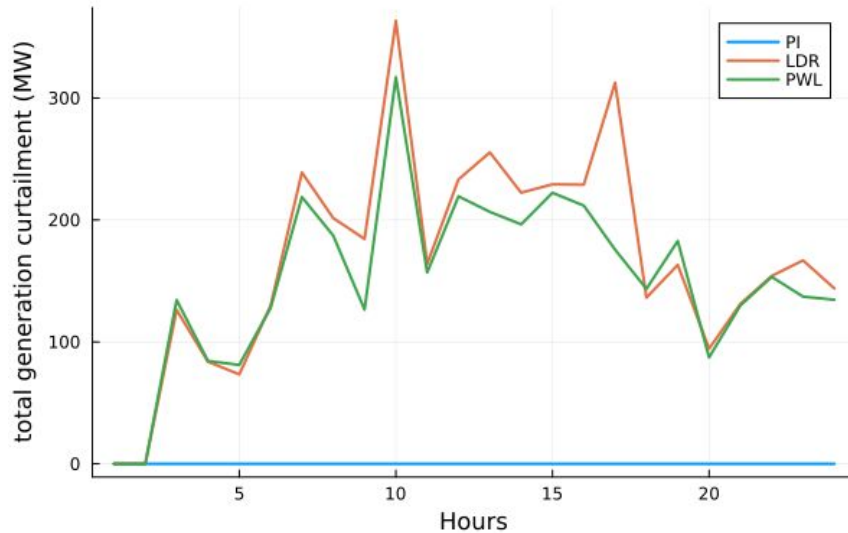


Total out of sample down-spinning reserve

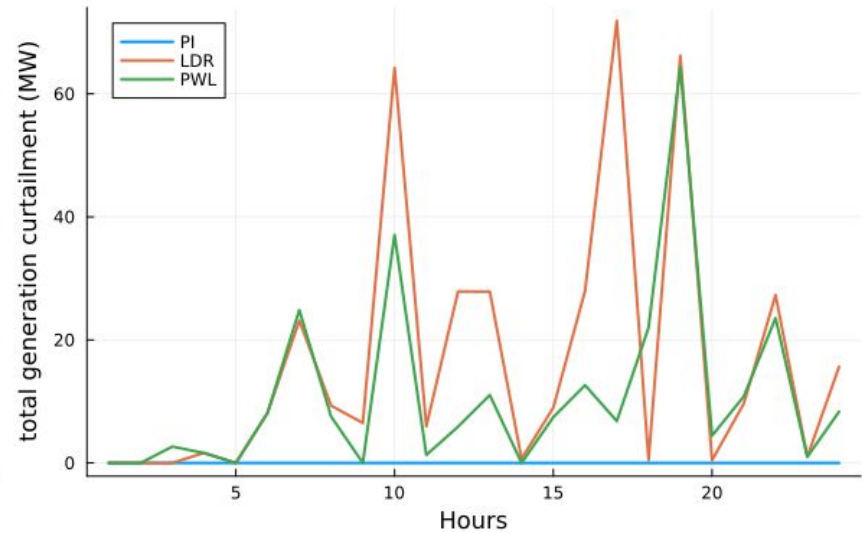


Results

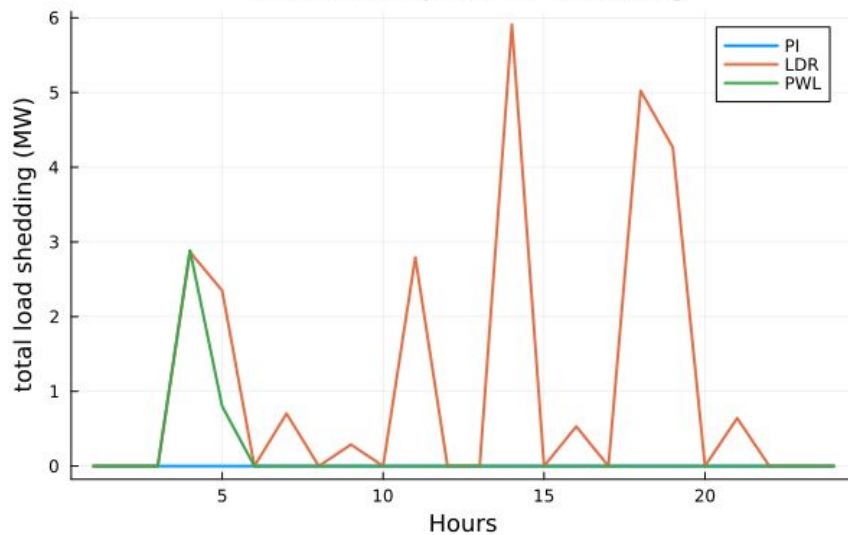
Total in sample generation curtailment



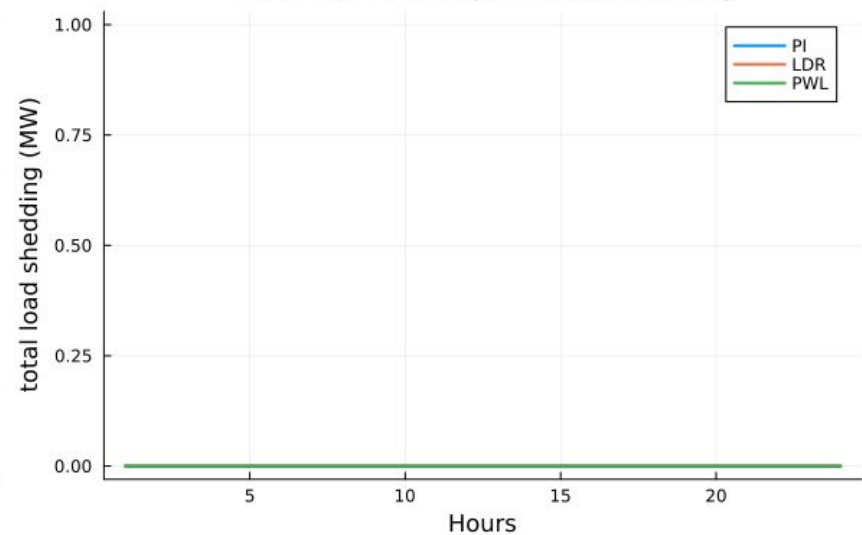
Total out of sample generation curtailment



Total in sample load shedding



Total out of sample load shedding



Conclusion

- PWL reduced both in sample and out of sample costs
 - Generation;
 - Reserve Down;
 - Curtailment;
 - Load-shedding (in sample);
- PWL increased computational time

- [1] Rodrigues, M., Street, A., & Arroyo, J. M. (2024). Multistage day-ahead scheduling of energy and reserves. *Electric Power Systems Research*, 235, 110793.